Network 1: Comprehensive Study Notes

These university-level study notes cover every topic in the Network 1 syllabus with rigorous academic depth, step-by-step worked examples, and exam-focused analysis. Designed for undergraduate students in computer science and engineering, the material builds from first principles to mastery — preparing you for Network 2 and Advanced Networking. Topics span network architecture, the OSI and TCP/IP models, physical layer theory, data link protocols, MAC mechanisms, IP addressing, subnetting, routing algorithms, and ICMP.

UNDERGRADUATE COMPUTER NETWORKS | ACADEMIC STUDY NOTES | NETWORK 1 → NETWORK 2

1. Introduction to Computer Networks

CHAPTER 1

1.1 What Is a Computer Network?

A **computer network** is a system of interconnected devices (nodes) that exchange data using shared protocols over transmission media. Networks enable resource sharing, communication, distributed processing, and centralized management.

PAN
Personal Area Network — up to 10 m. Bluetooth, USB.

LAN
Local Area Network — up to 1 km. Ethernet, Wi-Fi.

MAN
Metropolitan Area Network — up to 50 km. Fiber, WiMAX.

WAN
Wide Area Network — unlimited. MPLS, leased lines.

1.2 Network Architecture and Design Principles

Network architecture defines the physical and logical structure, protocols, and technologies used. Core principles:

Layering
Divide functionality into discrete layers, each serving the layer above.

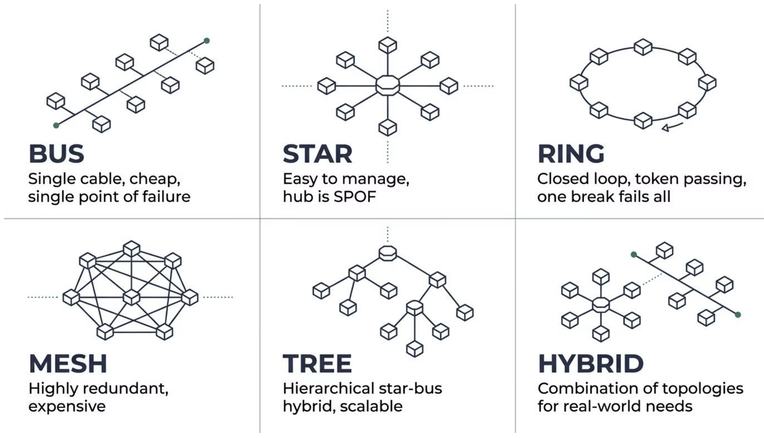
Modularity
Components can be replaced without affecting others.

Scalability
Network grows without redesign.

Fault Tolerance
Continues operating despite component failures.

Security
Confidentiality, integrity, and availability built in.

1.3 Network Topologies



Physical topology describes cable layout; **logical topology** describes data flow. Ethernet uses a physical star but a logical bus (broadcast domain). Modern switched networks use physical and logical star.

1.4 Network Components

Device	OSI Layer	Function	Key Fact
Hub	Layer 1 (Physical)	Broadcasts to all ports	Single collision domain; obsolete
Switch	Layer 2 (Data Link)	Forwards by MAC address	Per-port collision domain
Router	Layer 3 (Network)	Routes by IP address	Separates broadcast domains
Bridge	Layer 2	Connects two LAN segments	Filters by MAC; largely replaced
Access Point	Layer 2	Wireless to wired bridge	Uses CSMA/CA
Firewall	Layers 3–7	Filters traffic by policy	Stateful or stateless

1.5 Protocols and Standards

A **protocol** is a formal specification governing communication between devices — defining syntax, semantics, and timing. Key standards bodies:

IETF
Internet Engineering Task Force — TCP/IP, HTTP, DNS (RFCs).

IEEE
802.3 (Ethernet), 802.11 (Wi-Fi), 802.1Q (VLANs).

ITU-T
International Telecommunication Union — physical layer standards.

ISO
International Organization for Standardization — OSI model.

Exam Trap: A **protocol** defines the rules; a **standard** is a ratified specification. Not all protocols are standards, but all standards define protocols.

Section 1 Review

Key Concepts

- Network types: PAN, LAN, MAN, WAN
- Topology trade-offs: cost vs. redundancy
- Device layer mapping (hub L1, switch L2, router L3)
- Layering, modularity, fault tolerance

MCQs

1. Which device operates at Layer 2?
a) Hub b) Switch c) Repeater d) Modem
Answer: b

2. Most fault-tolerant topology?
a) Bus b) Star c) Ring d) Full Mesh
Answer: d

Conceptual Questions

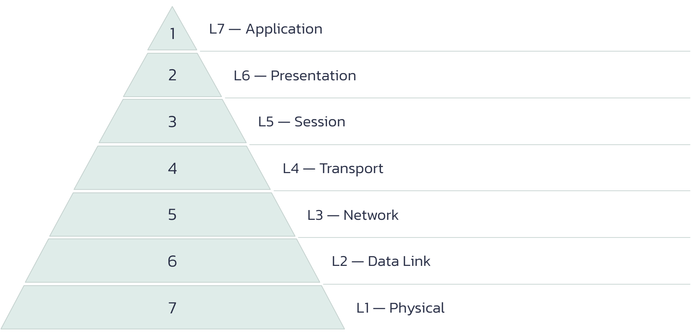
- Why is layering essential in network design?
- Compare physical vs. logical topology with examples.

2. Network Layers and Protocols

CHAPTER 2

2.1 The OSI Reference Model

The **Open Systems Interconnection (OSI)** model, defined by ISO in 1984, divides network communication into **seven layers**. Each layer provides services to the layer above and uses services from the layer below. The mnemonic **"Please Do Not Throw Sausage Pizza Away"** helps remember the order.



Layer	PDU	Key Protocols	Function
7 — Application	Data	HTTP, FTP, SMTP, DNS	User interface to network
6 — Presentation	Data	SSL/TLS, JPEG, ASCII	Data formatting, encryption
5 — Session	Data	NetBIOS, RPC, PPTP	Session management
4 — Transport	Segment (TCP) / Datagram (UDP)	TCP, UDP	End-to-end delivery, reliability
3 — Network	Packet	IP, ICMP, OSPF	Logical addressing, routing
2 — Data Link	Frame	Ethernet, 802.11, PPP	Physical addressing, error detection
1 — Physical	Bits	RS-232, 802.3 physical	Bit transmission over medium

2.2 TCP/IP Protocol Suite

The **TCP/IP model** (also called the Internet model) condenses OSI into four layers. It is the practical foundation of the modern Internet.

OSI Model (7 Layers)

1. Application
2. Presentation
3. Session
4. Transport
5. Network
6. Data Link
7. Physical

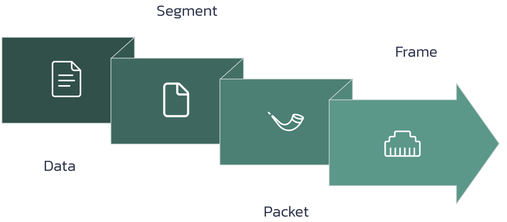
TCP/IP Model (4 Layers)

1. Application (OSI L5–L7)
2. Transport (OSI L4)
3. Internet (OSI L3)
4. Network Access (OSI L1–L2)

OSI Layers 5–7 map to TCP/IP Application. OSI Layers 1–2 map to TCP/IP Network Access.

2.3 Encapsulation and Decapsulation

Encapsulation occurs at the sender: each layer wraps data with a header (and sometimes trailer). **Decapsulation** reverses this at the receiver.



Each layer's PDU has a distinct name: **Data → Segment → Packet → Frame → Bits**. Headers are stripped in reverse order at the destination.

2.4 Addressing and Naming Schemes

Address Type	Layer	Format	Example	Scope
MAC Address	Data Link (L2)	48-bit hex	00:1A:2B:3C:4D:5E	Local (LAN)
IPv4 Address	Network (L3)	32-bit dotted decimal	192.168.1.1	Global
IPv6 Address	Network (L3)	128-bit hex	2001:db8::1	Global
Port Number	Transport (L4)	16-bit integer	80 (HTTP), 443 (HTTPS)	Host-local
URL / URI	Application (L7)	Text string	https://www.example.com/path	Global

A **socket** = IP address + port number, uniquely identifying an endpoint. **FQDN** (Fully Qualified Domain Name) is resolved to IP via DNS.

📚 Section 2 Review

Key Concepts

- OSI 7 layers with PDU names and functions
- TCP/IP 4-layer mapping to OSI
- Encapsulation: Data → Segment → Packet → Frame → Bits
- Address types: MAC (L2), IP (L3), Port (L4)

MCQs

1. TCP/IP Transport maps to which OSI layer?
a) L3 b) L4 c) L5 d) L7
Answer: b

2. PDU at Network layer?
a) Frame b) Segment c) Packet d) Bit
Answer: c

Conceptual Questions

- Trace encapsulation for an HTTP request from browser to server.
- Why does TCP/IP merge OSI Layers 5–7?

3. Physical Layer and Transmission Media

CHAPTER 3

3.1 Data Transmission Concepts

The **Physical Layer (L1)** transmits raw bits over a medium. Key parameters:

Simplex

One direction only. Example: Radio broadcast.

Half-Duplex

Both directions, not simultaneously. Example: Walkie-talkie.

Full-Duplex

Simultaneous bidirectional. Example: Ethernet, telephone.

Serial vs. Parallel

Serial: one bit at a time (modern networks). Parallel: multiple bits simultaneously (legacy).

Synchronous vs. Asynchronous

Synchronous: shared clock. Asynchronous: start/stop bits per character.

3.2 Analog vs. Digital Transmission

Analog Transmission

Continuous signal varying over time. Susceptible to noise; amplified signals amplify noise too.

- Example: AM/FM radio, PSTN voice
- Measured in Hertz (Hz)
- Modulation: AM, FM, PM

Digital Transmission

Discrete signal (0s and 1s). Noise-resistant; regenerators restore clean signals.

- Example: Ethernet, fiber optics
- Measured in bits per second (bps)
- Encoding: NRZ, Manchester, 4B/5B

3.3 Transmission Media

Media	Type	Bandwidth	Max Distance	Immunity to EMI	Cost
UTP Cat5e	Guided (Copper)	1 Gbps	100 m	Low	Low
UTP Cat6	Guided (Copper)	10 Gbps	55 m	Low	Medium
Coaxial	Guided (Copper)	10 Mbps–1 Gbps	500 m	Medium	Medium
Multi-mode Fiber	Guided (Optical)	10 Gbps+	2 km	Excellent	High
Single-mode Fiber	Guided (Optical)	100 Gbps+	40+ km	Excellent	Very High
Wi-Fi (2.4 GHz)	Unguided (Wireless)	54–600 Mbps	~100 m	N/A	Low

3.4 Signal Encoding

Encoding converts digital data to signals for transmission.

NRZ (Non-Return to Zero)

High = 1, Low = 0. Problem: long runs cause clock drift. No error detection.

Manchester Encoding

Bit-period transition: low→high = 1, high→low = 0. Self-clocking. Used in 10BASE-T Ethernet. Efficiency: 50%.

4B/5B Encoding

4 data bits → 5-bit code. Guarantees transitions. Used in Fast Ethernet (100BASE-TX). Efficiency: 80%.

3.5 Attenuation, Noise, and Channel Capacity

Attenuation is signal power loss over distance, measured in decibels:

$$dB = 10 \log_{10} \left(\frac{P_{out}}{P_{in}} \right)$$

A –3 dB loss means 50% power remaining. **Noise** types: thermal, intermodulation, crosstalk, impulse.

Signal-to-Noise Ratio (SNR):

$$SNR_{dB} = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right)$$

Nyquist Theorem (Noiseless Channel)

$$C = 2B \log_2 M$$

Where C = capacity (bps), B = bandwidth (Hz), M = signal levels.

> **Worked Example (Nyquist):** Noiseless channel, B = 3000 Hz, M = 4 levels.
> $C = 2 \times 3000 \times \log_2 4 = 2 \times 3000 \times 2 = 12{,}000$ bps = **12 kbps**

Shannon Capacity (Noisy Channel)

$$C = B \log_2(1 + SNR)$$

Where SNR is linear (not dB). Convert: $SNR = 10^{(SNR_{dB}/10)}$.

> **Worked Example (Shannon):** B = 3000 Hz, SNR = 30 dB.
> $SNR = 10^3 = 1000$
> $C = 3000 \times \log_2(1001) \approx 3000 \times 9.97 \approx 29{,}900$ bps ≈ **30 kbps**

3.6 Delay, Throughput, and Bandwidth

Concept	Definition	Formula
Transmission Delay	Time to push all bits onto wire	$d_{trans} = L/R$
Propagation Delay	Time for bit to travel medium	$d_{prop} = d/s$
Processing Delay	Time to examine packet header	Typically microseconds
Queuing Delay	Time waiting in router buffer	Variable (congestion-dependent)
Total Delay	Sum of all delays	$d_{total} = d_{trans} + d_{prop} + d_{proc} + d_{queue}$
Throughput	Actual data rate achieved	$\text{Throughput} \leq \min(\text{link capacities})$
Bandwidth-Delay Product	Bits "in flight" on the link	$BDP = R \times d_{prop}$

> **Worked Example (Delay):** L = 10,000 bits, R = 1 Mbps, distance = 2,000 km, speed = 2×10^8 m/s.
> $d_{trans} = 10{,}000 / 10^6 = 10$ ms
> $d_{prop} = 2 \times 10^6 / 2 \times 10^8 = 10$ ms
> **Total = 20 ms** (ignoring processing/queuing)

📚 Section 3 Review

Key Formulas

- Nyquist: $C = 2B \log_2 M$
- Shannon: $C = B \log_2(1 + SNR)$
- SNR dB: $10 \log_{10}(P_{signal}/P_{noise})$
- Transmission delay: L/R
- Propagation delay: d/s

Exam Traps

- Shannon uses linear SNR, not dB — convert first
- Bandwidth ≠ Throughput (throughput ≤ bandwidth)
- Manchester efficiency is 50%, not 100%

Practice Problems

1. Channel B = 4 kHz, SNR = 63 (linear). Find C.
Answer: C = 4000 × log₂(64) = 4000 × 6 = 24 kbps

2. File = 5 Mb, link = 10 Mbps, distance = 5000 km, speed = 2.5×10^8 m/s. Find total delay.
Answer: d_trans = 0.5 s, d_prop = 0.02 s, total = 0.52 s

4. Data Link Layer and Medium Access Control

CHAPTER 4

4.1 Data Link Layer Functions

The **Data Link Layer (L2)** provides node-to-node delivery across a single link. Functions:

1. **Framing**
   Encapsulates Network layer packets into frames with headers and trailers.
2. **Physical Addressing**
   Uses 48-bit MAC addresses in frame headers.
3. **Flow Control**
   Prevents fast sender from overwhelming slow receiver.
4. **Error Detection**
   Detects bit errors via CRC or parity in the frame trailer.
5. **Medium Access Control**
   Controls when a node transmits on a shared medium.

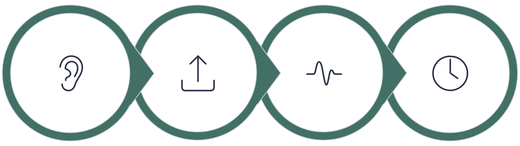
4.2 Ethernet Frame Structure (IEEE 802.3)

Field	Size	Purpose
Preamble	7 bytes	Synchronization pattern (10101010...)
SFD (Start Frame Delimiter)	1 byte	Marks frame start (10101011)
Destination MAC	6 bytes	Receiver's MAC address
Source MAC	6 bytes	Sender's MAC address
Length / Type	2 bytes	Payload length or protocol type
Data (Payload)	46–1500 bytes	Encapsulated packet (IP)
FCS (Frame Check Sequence)	4 bytes	CRC-32 for error detection

Minimum frame size = 64 bytes (enforces collision detection in legacy half-duplex Ethernet). Maximum = 1518 bytes (standard) or 1522 with 802.1Q VLAN tag.

4.3 MAC Protocols: Wired (CSMA/CD)

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is used in legacy half-duplex Ethernet (IEEE 802.3).



Listen Transmit Monitor Backoff

Binary Exponential Backoff: After n collisions, wait a random time from $[0, 2^k - 1]$ slot times, where $k = \min(n, 10)$. After 16 collisions, the frame is discarded.

> ⚠ **Important:** CSMA/CD is obsolete in modern full-duplex switched Ethernet. Switches create point-to-point links with no collisions. CSMA/CD appears in exams for historical context.

4.4 MAC Protocols: Wireless (CSMA/CA)

CSMA/CA (Collision Avoidance) is used in Wi-Fi (IEEE 802.11). Collisions cannot be detected wirelessly due to the **hidden node problem**.

- → **Sense the Channel**
  If busy, defer. If idle for DIFS, proceed.
- → **Random Backoff**
  Choose random slots from contention window. Counter decrements only when channel is idle.
- → **Transmit**
  Send frame when counter reaches zero.
- → **Wait for ACK**
  Receiver sends ACK after SIFS. No ACK = collision; retransmit.

RTS/CTS (Request to Send / Clear to Send) solves the hidden node problem by reserving the channel before data transmission.

4.5 CSMA/CD vs. CSMA/CA — Comparison

Feature	CSMA/CD	CSMA/CA
Medium	Wired (Ethernet)	Wireless (Wi-Fi)
Strategy	Detect collisions during transmission	Avoid collisions before transmission
Collision Handling	Jam signal + Binary Exponential Backoff	Random backoff + ACK timeout
ACK Required	No	Yes (every unicast frame)
Hidden Node Problem	Not applicable	Solved by RTS/CTS
Efficiency	High in wired LANs	Lower due to overhead
Standard	IEEE 802.3	IEEE 802.11

📚 Section 4 Review

Key Concepts

- Five Data Link functions: framing, addressing, flow control, error detection, MAC
- Ethernet frame fields and sizes
- CSMA/CD vs. CSMA/CA differences
- Hidden node problem and RTS/CTS solution

MCQs

1. Which MAC protocol uses ACK frames?
a) CSMA/CD b) CSMA/CA c) Token Ring d) Aloha
Answer: b

2. Ethernet frame FCS field size?
a) 2 bytes b) 4 bytes c) 6 bytes d) 8 bytes
Answer: b

Conceptual Questions

- Why can't wireless networks use collision detection?
- Explain the hidden node problem with a diagram.

5. Error Detection and Correction

CHAPTER 4 — CONTINUED

5.1 Types of Transmission Errors

Errors occur due to noise, interference, and attenuation.

Single-Bit Error
Exactly one bit flips. Rare in serial transmission.

Burst Error
Multiple consecutive bits corrupted. Common in practice. Length = first to last corrupted bit.

5.2 Parity Check

A single parity bit detects all **odd-numbered** bit errors but cannot detect even-numbered errors or correct any errors.

Even Parity

Parity bit set so total 1s is even.

Data: `1011001` (four 1s) → Parity = `0`
Data: `1011011` (five 1s) → Parity = `1`

Odd Parity

Parity bit set so total 1s is odd.

Data: `1011001` (four 1s) → Parity = `1`
Data: `1011011` (five 1s) → Parity = `0`

5.3 Checksum

The sender divides data into n-bit segments, sums them using **1's complement arithmetic**, and sends the complemented sum. The receiver recomputes; a result of all 1s indicates no error. Used in IP, TCP, and UDP headers.

5.4 Cyclic Redundancy Check (CRC)

CRC is the most powerful error-detection method in data communications. The sender and receiver agree on a **generator polynomial G(x)**. The sender appends r zero bits (where r = degree of G), divides using **modulo-2 arithmetic** (XOR), and appends the remainder (FCS). The receiver divides the received frame by G; zero remainder = no error.

Worked Example (CRC): Data M = `110101`, Generator G = `1011` (degree r = 3).
Step 1: Append 3 zeros → `110101000`
Step 2: Modulo-2 divide `110101000` ÷ `1011`
Step 3: Remainder = `001` (FCS)
Transmitted frame: `110101001`
Receiver divides `110101001` ÷ `1011` → remainder `0` = no error

CRC-32 (IEEE 802.3) detects: all single-bit errors, all double-bit errors, all odd-bit errors, all burst errors ≤ 32 bits, and most longer burst errors.

5.5 Hamming Code (Error Correction)

Hamming codes detect and **correct** single-bit errors. Parity bits are placed at positions that are powers of 2 (1, 2, 4, 8, ...). For m data bits and r parity bits: $2^r \geq m + r + 1$.

Worked Example (Hamming): Encode data `1011` (m = 4). Need r where $2^r \geq 4 + r + 1$ → r = 3.
Positions: P1, P2, D1, P4, D2, D3, D4
Data placement: _ _ 1 _ 0 1 1
P1 (positions 1,3,5,7): covers bits 1,1,0,1 → even parity → P1 = `1`
P2 (positions 2,3,6,7): covers bits 1,0,1,1 → even parity → P2 = `1`
P4 (positions 4,5,6,7): covers bits 0,0,1,1 → even parity → P4 = `0`
Encoded: `1110011`
If bit 5 flips → received `1110111`
Syndrome: P1 check = 1, P2 check = 0, P4 check = 1 → Syndrome = 101 = position 5 → correct bit 5

5.6 Error Detection/Correction Comparison

Method	Detects	Corrects	Overhead	Used In
Parity	Odd-bit errors only	No	1 bit	Memory (RAM)
Checksum	Most errors	No	16 bits	IP, TCP, UDP
CRC	All burst ≤ r bits	No	r bits (e.g., 32)	Ethernet, Wi-Fi
Hamming Code	Single-bit errors	Single-bit errors	$\log_2(n)$ bits	ECC Memory
Reed-Solomon	Burst errors	Burst errors	High	CDs, DVDs, QR codes

📚 Section 5 Review

Key Formulas

- Hamming condition: $2^r \geq m + r + 1$
- CRC: $T(x) = M(x) \cdot x^r + R(x)$

Exam Traps

- Parity cannot detect even-numbered bit errors
- CRC detects but does NOT correct errors
- Hamming corrects single-bit; cannot correct burst errors

Practice Problems

1. How many parity bits for 16 data bits (Hamming)?
Answer: 2^r ≥ 16+r+1 → r = 5 (since 2^5 = 32 ≥ 22)

2. CRC: M = 1010, G = 1011. Find FCS.
Answer: Append 000 → 1010000 ÷ 1011 → Remainder = 111. Frame = 1010111

6. Network Layer, IP Addressing, and Subnetting

CHAPTER 5

6.1 Network Layer Functions

The **Network Layer (L3)** provides host-to-host delivery across multiple networks. Core functions:

Logical Addressing

Assigns IP addresses independent of hardware. Enables global routing.

Routing

Selects optimal paths through the network using routing algorithms.

Forwarding

Moves packets from input to output port based on forwarding table.

Congestion Control

Manages traffic load to prevent router overload.

6.2 IPv4 Addressing

An **IPv4 address** is a 32-bit number in dotted-decimal notation, divided into **network** and **host** portions. IPv4 supports approximately 4.3 billion addresses.

IPv4 Address Classes

Class	Range	Network Bits	Networks	Hosts/Net
A	1–126	8	126	16.7 million
B	128–191	16	16,384	65,534
C	192–223	24	2.1 million	254
D	224–239	—	Multicast	—
E	240–255	—	Reserved	—

Special IPv4 Addresses

- `0.0.0.0` — Default route / this host
- `127.0.0.1` — Loopback (localhost)
- `255.255.255.255` — Limited broadcast
- `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16` — Private (RFC 1918)
- `169.254.0.0/16` — APIPA (auto-configured)

Private addresses are not routable on the Internet. NAT translates them to a public IP.

6.3 Subnetting

Subnetting divides a network into smaller subnets, reducing broadcast domains and improving efficiency. The subnet mask identifies network vs. host bits.

Worked Example (Subnetting): Network: 192.168.1.0/24. Need 4 subnets.
Step 1: Borrow bits — $2^2 = 4$ subnets → borrow 2 bits
Step 2: New prefix = /26
Step 3: Subnet mask = 255.255.255.192
Step 4: Hosts per subnet = $2^6 - 2 = 62$
Subnets: .0, .64, .128, .192 (each with 62 usable hosts)

6.4 CIDR and VLSM

CIDR (Classless Inter-Domain Routing) replaces classful addressing with flexible prefix lengths (e.g., /20), enabling route aggregation. **VLSM** (Variable Length Subnet Masking) allows different prefix lengths within the same network, optimizing address allocation.

Worked Example (VLSM): Network 192.168.1.0/24. Requirements: Subnet A = 50 hosts, Subnet B = 20 hosts, Subnet C = 10 hosts.
Subnet A: /26 (62 hosts) → 192.168.1.0/26
Subnet B: /27 (30 hosts) → 192.168.1.64/27
Subnet C: /28 (14 hosts) → 192.168.1.96/28
Remaining: 192.168.1.112/28 (reserved)

6.5 IPv6

IPv6 uses 128-bit addresses in hexadecimal, supporting 3.4×10^{38} addresses. Format: eight groups of four hex digits (`2001:0db8:85a3:0000:0000:8a2e:0370:7334`). Consecutive zeros compress to `::` (once per address).

Feature	IPv4	IPv6
Address Size	32 bits	128 bits
Address Notation	Dotted decimal	Hexadecimal
Total Addresses	~4.3 billion	3.4×10^{38}
Header Checksum	Yes	No (removed for speed)
Fragmentation	Sender and router	Sender only
Broadcast	Yes	No (uses multicast)
Security (IPsec)	Optional	Built-in
Address Configuration	DHCP	SLAAC or DHCPv6

📚 Section 6 Review

Key Formulas

- Subnets: 2^n (n = borrowed bits)
- Hosts per subnet: $2^h - 2$ (h = host bits)
- VLSM: allocate largest subnet first

Exam Traps

- Subnet zero and all-ones subnet are usable in modern networks (Cisco default)
- First and last addresses in a subnet are reserved (network + broadcast)
- CIDR /30 = 2 usable hosts (point-to-point links)

Practice Problems

1. Network 172.16.0.0/16. Create 8 subnets. Find new prefix, mask, hosts/subnet.
Answer: /19, 255.255.224.0, 2046 hosts/subnet

2. Summarize 192.168.0.0/24 through 192.168.7.0/24.
Answer: 192.168.0.0/21 (8 contiguous /24s = 2^3, so 24–3 = 21)

7. Routing Algorithms and Protocols

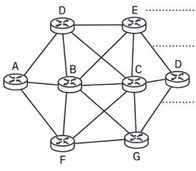
CHAPTER 5 — CONTINUED

7.1 Routing vs. Forwarding

Forwarding is the local action of moving a packet from an input port to an output port using a **forwarding table**. **Routing** is the global process of determining optimal paths using **routing algorithms** and populating the forwarding table.

7.2 Routing Algorithm Classification

Global Routing Algorithms (Link-State)



Dijkstra's algorithm

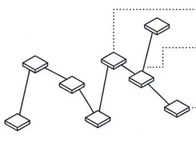
Full network topology known

Shortest path computed independently by each router

Examples: OSPF, IS-IS

- **Convergence Speed:** Fast
- **Message Complexity:** High
- **Robustness:** High

Decentralized Routing Algorithms (Distance-Vector)



Bellman-Ford algorithm

Only knows neighbor costs

Iterative, asynchronous computation

Examples: RIP, IGRP

- **Convergence Speed:** Slow
- **Message Complexity:** Low
- **Robustness:** Low

7.3 Dijkstra's Algorithm (Link-State)

Each router floods **Link-State Advertisements (LSAs)** to all routers, building a complete topology map. Dijkstra's algorithm computes the shortest path tree from source to all destinations.

01

Initialize

Set D(source) = 0, D(all others) = ∞. N' = {source}.

02

Select Minimum

Find node w not in N' with minimum D(w). Add w to N'.

03

Update Distances

For each neighbor v of w: D(v) = min(D(v), D(w) + c(w,v)).

04

Repeat

Repeat steps 2–3 until all nodes are in N'.

Worked Example (Dijkstra): Graph: A-B=2, A-C=5, B-C=1, B-D=4, C-D=2. Source = A.
Init: D(A)=0, D(B)=∞, D(C)=∞, D(D)=∞. N'={A}
Iter 1: Select B (D=2). Update C: min(∞, 2+1)=3. Update D: min(∞, 2+4)=6. N'={A,B}
Iter 2: Select C (D=3). Update D: min(6, 3+2)=5. N'={A,B,C}
Iter 3: Select D (D=5). N'={A,B,C,D}
Shortest paths: A→B=2, A→C=3 (via B), A→D=5 (via B→C)

7.4 Bellman–Ford Algorithm (Distance-Vector)

Each router maintains a **distance vector** — its estimated cost to every destination. Routers exchange vectors only with direct neighbors. The **Bellman-Ford equation**:

$$D_x(y) = \min_v \{c(x,v) + D_v(y)\}$$

Where $D_x(y)$ = cost from x to y, $c(x,v)$ = direct link cost, $D_v(y)$ = neighbor's cost to y.

⚠ **Count-to-Infinity Problem:** When a link fails, bad news propagates slowly. Routers may increment costs indefinitely. Solutions: **Split Horizon** (don't advertise route back to the neighbor it came from), **Poison Reverse** (advertise infinite cost back), and **Triggered Updates** (send updates immediately on change).

7.5 Routing Protocol Comparison

Protocol	Type	Algorithm	Metric	Max Hop Count	Convergence	Standard
RIPv1	Distance-Vector	Bellman-Ford	Hop count	15	Slow	Proprietary (original)
RIPv2	Distance-Vector	Bellman-Ford	Hop count	15	Slow	RFC 2453
EIGRP	Advanced DV	DUAL	Bandwidth + Delay	224	Fast	Cisco (partial open)
OSPF	Link-State	Dijkstra	Cost (inverse BW)	None	Fast	RFC 2328
IS-IS	Link-State	Dijkstra	Cost	None	Fast	ISO/IEC 10589
BGP	Path-Vector	Policy-based	Attributes	None	Slow	RFC 4271

OSPF is the most common interior gateway protocol (IGP) in enterprise networks. **BGP** is the exterior gateway protocol (EGP) used between autonomous systems on the Internet.

📚 Section 7 Review

Key Concepts

- Dijkstra's algorithm: global knowledge, link-state, OSPF
- Bellman-Ford: neighbor knowledge, distance-vector, RIP
- Count-to-infinity problem and solutions
- Routing protocol comparison by type, metric, convergence

MCQs

1. Which protocol uses Dijkstra's algorithm?
a) RIP b) EIGRP c) OSPF d) BGP
Answer: c

2. RIP maximum hop count?
a) 16 b) 15 c) 255 d) Unlimited
Answer: b (15 is reachable; 16 = infinity)

Conceptual Questions

- Compare link-state and distance-vector on convergence, overhead, and scalability.
- Explain how split horizon prevents routing loops.

8. ICMP and End-of-Syllabus Review

CHAPTER 5 — FINAL TOPIC

8.1 ICMP — Internet Control Message Protocol

ICMP (Internet Control Message Protocol) operates at the Network Layer (L3) and is used for diagnostic and control purposes. ICMP messages are encapsulated within IP packets (IP protocol number **1**). ICMP does **not** make IP reliable — it only reports errors.

Type	Code	Message Name	Purpose
0	0	Echo Reply	Response to ping
3	0–15	Destination Unreachable	Network, host, port, or protocol unreachable
4	0	Source Quench	Congestion control (deprecated)
5	0–3	Redirect	Tell host to use a better gateway
8	0	Echo Request	Ping request
11	0	Time Exceeded (TTL)	TTL reached zero (used by traceroute)
11	1	Time Exceeded (Reassembly)	Fragment reassembly timeout
12	0	Parameter Problem	Invalid IP header field

8.2 Ping and Traceroute

Ping (ICMP Echo)

Sends ICMP Type 8 (Echo Request) to a target. Target replies with Type 0 (Echo Reply). Measures **Round-Trip Time (RTT)** and confirms reachability.

```
ping 8.8.8.8
```

- RTT = transmission + propagation + processing + queuing (×2)
- No reply = host down, firewall blocking, or network unreachable

Traceroute (ICMP TTL Expiry)

Discovers the path to a destination by exploiting TTL (Time To Live).

1. Send packet with TTL=1 → first router decrements to 0, discards, sends Type 11 (Time Exceeded)
2. Send packet with TTL=2 → second router replies
3. Repeat until destination responds with Echo Reply

Windows uses ICMP; Unix/Linux traceroute uses UDP by default.

8.3 ICMPv6

ICMPv6 (RFC 4443) extends ICMP for IPv6 and incorporates functionality that was separate in IPv4. Key additions:

Neighbor Discovery (ND)

Replaces ARP. Discovers link-layer addresses of neighbors on the same link.

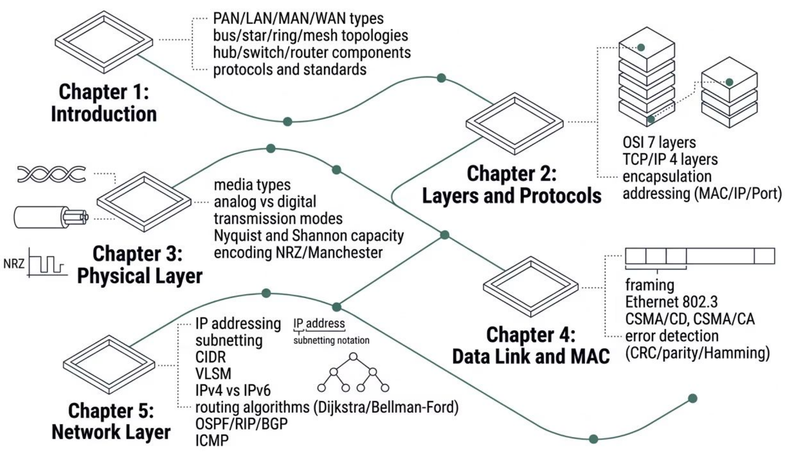
Router Discovery

Hosts discover local routers via Router Advertisement (RA) and Router Solicitation (RS) messages.

SLAAC

Stateless Address Autoconfiguration — hosts generate their own IPv6 addresses using RA prefixes.

8.4 End-of-Syllabus: Complete Topic Map



📚 Section 8 Review

Key Concepts

- ICMP Type 8/0 = Echo Request/Reply (ping)
- ICMP Type 3 = Destination Unreachable
- ICMP Type 11 = Time Exceeded (traceroute)
- ICMP is encapsulated in IP (protocol 1)
- ICMPv6 includes Neighbor Discovery and SLAAC

MCQs

1. ICMP Type 11 indicates:
a) Echo Reply b) Destination Unreachable c) Time Exceeded d) Redirect
Answer: c

2. Which IPv6 protocol replaces ARP?
a) DHCPv6 b) Neighbor Discovery c) SLAAC d) ICMPv4
Answer: b

Conceptual Questions

- Explain how traceroute uses TTL and ICMP Time Exceeded messages.
- Why does ICMP not make IP reliable?

Network 1: Complete Exam Preparation

FINAL REVIEW

10.1 Master Comparison Tables

TCP vs. UDP

Feature	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Guaranteed delivery, ACKs, retransmission	Best-effort, no ACKs
Ordering	Guaranteed in-order delivery	No ordering guarantee
Flow Control	Sliding window	None
Congestion Control	Yes (AIMD, slow start)	None
Overhead	High (20-byte header)	Low (8-byte header)
Speed	Slower	Faster
Use Cases	HTTP, HTTPS, FTP, SSH, SMTP	DNS, DHCP, VoIP, streaming

Analog vs. Digital Transmission

Feature	Analog	Digital
Signal Type	Continuous waveform	Discrete levels (0 and 1)
Noise Susceptibility	High — noise accumulates	Low — regenerators clean signal
Amplification	Amplifiers (amplify noise too)	Regenerators (restore clean signal)
Bandwidth Usage	Typically lower	Typically higher
Examples	AM/FM radio, PSTN voice	Ethernet, fiber optics, Wi-Fi

Bandwidth vs. Throughput vs. Latency

Concept	Definition	Unit	Analogy
Bandwidth	Maximum theoretical data rate of a link	bps, Mbps, Gbps	Width of a pipe
Throughput	Actual data rate achieved (≤ bandwidth)	bps, Mbps, Gbps	Water actually flowing
Latency	Time for a packet to travel source to destination	Milliseconds (ms)	Time for water to travel
Goodput	Useful data rate (excludes headers/retransmissions)	bps, Mbps, Gbps	Usable water delivered

10.2 Critical Formulas Reference Sheet

Physical Layer

- Nyquist: $C = 2B \log_2 M$
- Shannon: $C = B \log_2(1 + SNR)$
- SNR (dB): $10 \log_{10}(P_{signal}/P_{noise})$
- Transmission delay: $d_{trans} = L/R$
- Propagation delay: $d_{prop} = d/s$
- BDP: $R \times d_{prop}$

Data Link Layer

- Hamming: $2^r \geq m + r + 1$
- CRC: $T(x) = M(x) \cdot x^r + R(x)$
- Ethernet min frame: 64 bytes
- Ethernet max frame: 1518 bytes

Network Layer

- Subnets: 2^n
- Hosts/subnet: $2^h - 2$
- IPv4 total: $2^{32} \approx 4.3$ billion
- IPv6 total: $2^{128} \approx 3.4 \times 10^{38}$